JHU Engineering for Professionals
Applied and Computational Mathematics
Data Mining: 625.740 Fall '26

Midterm Exam

Please type up your answers. If you are not able to type your answers, please write very neatly. Sloppy, small, or otherwise difficult-to-read writing will not be accepted. Please include in your submission all code in electronic form.

You may use the textbook Pattern Classification, by Duda, et al. and other textbooks listed in the course syllabus. You may use the material posted to Canvas and Piazza: notes, videos, textbook chapters. Please cite *all* sources. No other sources may be consulted during this exam.

Posting questions to or viewing questions at "homework and exam" solving websites and the like is not allowed. Searching the internet, or obtaining notes from other classes is not allowed on this exam. During the time of the exam, you may discuss these problems only with the course instructors and nobody else.

"GenAI" may not be used in any form. This includes ChatGPT and all of its ilk and Overleaf AI Assist. Students must complete all work independently without GenAI assistance, including drafting, editing, or revision. Use of GenAI on this or any exam constitutes a violation of academic integrity.

Name: _____

Time and date of exam start: _____

Time and date of exam finish: _____

This exam represents my own work. I did not consult with anybody or access any prohibited sources during the time of the exam.

Signature: _____ **Date:** _____

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1. Recall that if $f(\mathbf{x})$ and $g(\mathbf{x})$ are probability distributions, then the Kullback-Leibler divergence of g from f is

$$D(f,g) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(\mathbf{x}) \log \left[\frac{f(\mathbf{x})}{g(\mathbf{x})} \right] d\mathbf{x}.$$

   We seek to approximate an arbitrary distribution $f(\mathbf{x})$ by a normal distribution $g(\mathbf{x}) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Show that the values that minimize $D(f,g)$ are

$$\boldsymbol{\mu} = \mathbb{E}_{f(\mathbf{x})}[\mathbf{x}],$$
$$\Sigma = \mathbb{E}_{f(\mathbf{x})}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T].$$

2. The columns in the file `midterm_exam_data.txt` are the X and Y values of a function which is a sum of cosines with added Gaussian noise.

   (a) For various values of n, fit an nth order cosine series ($y = \frac{a_0}{2} + \sum_{k=1}^{n} a_k \cos kx$) to the data. What would you say is the most likely value of n?

   (b) Estimate the level of the noise.

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3. Consider a Bayesian classifier, with distributions being n-dimensional multivariate Bernoulli random variables. The conditional probability for a given category is

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i}(1-\theta_i)^{1-x_i},$$

and let $\mathcal{D} = \{\mathbf{x}_1, \cdots, \mathbf{x}_k\}$ be a set of k samples independently drawn from this probability density.

(a) If $\mathbf{s} = (s_1, \cdots, s_n)^T$ is the sum of the k samples, show that

$$P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

(b) Assuming a uniform prior distribution for $\boldsymbol{\theta}$ show that

$$P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

(c) Plot this density for the case $n = 1, k = 1$ and for the two resulting possibilities for s_i.

(d) Integrate the product $P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})$ over $\boldsymbol{\theta}$ to obtain

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} \left(\frac{s_i+1}{k+2}\right)^{x_i} \left(1-\frac{s_i+1}{k+2}\right)^{1-x_i}$$

(e) Compare the expressions for $P(\mathbf{x}|\mathcal{D})$ and $P(\mathbf{x}|\boldsymbol{\theta})$. What is $\hat{\boldsymbol{\theta}}$, the effective Bayesian estimate for $\boldsymbol{\theta}$?